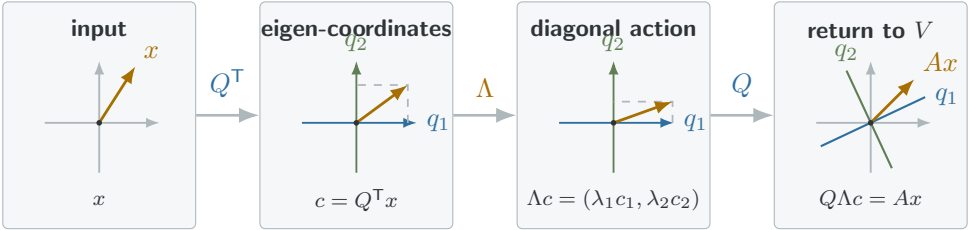
orthogonal diagonalization

for a real symmetric matrix $A = A^{\mathsf{T}}$



$$A = Q\Lambda Q^{\mathsf{T}} \qquad Q^{\mathsf{T}}Q = I \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

the columns $q_1, \ldots, q_n$ of Q are orthonormal eigenvectors of A